

Aim : To Develop regression model and evaluate its performance

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```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

Import File

```
In [2]: ap=pd.read_csv(r"C:\Users\admin\Downloads\advertising.csv")
```

Visualize data

```
In [3]: ap.head()
```

```
Out[3]:
```

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

```
In [4]: ap.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0    TV          200 non-null    float64
1    Radio       200 non-null    float64
2    Newspaper   200 non-null    float64
3    Sales       200 non-null    float64
dtypes: float64(4)
memory usage: 6.4 KB
```

```
In [5]: ap.describe()
```

```
Out[5]:
```

	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	147.042500	23.264000	30.554000	15.130500
std	85.854236	14.846809	21.778621	5.283892
min	0.700000	0.000000	0.300000	1.600000
25%	74.375000	9.975000	12.750000	11.000000
50%	149.750000	22.900000	25.750000	16.000000
75%	218.825000	36.525000	45.100000	19.050000
max	296.400000	49.600000	114.000000	27.000000

```
In [6]: ap.shape
```

```
Out[6]: (200, 4)
```

```
In [7]: ap.isnull().sum()
```

```
Out[7]: TV          0
Radio          0
Newspaper      0
Sales          0
dtype: int64
```

```
In [8]: ap.tail()
```

```
Out[8]:
```

	TV	Radio	Newspaper	Sales
195	38.2	3.7	13.8	7.6
196	94.2	4.9	8.1	14.0
197	177.0	9.3	6.4	14.8
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	18.4

Feature Extraction

```
In [9]: x= ap.iloc[:, :-1]
```

```
In [10]: y =ap.iloc[:, -1]
```

```
In [11]: x.head()
```

```
Out[11]:
```

	TV	Radio	Newspaper
0	230.1	37.8	69.2
1	44.5	39.3	45.1
2	17.2	45.9	69.3
3	151.5	41.3	58.5
4	180.8	10.8	58.4

```
In [12]: y.head()
```

```
Out[12]: 0    22.1
1    10.4
2    12.0
3    16.5
4    17.9
Name: Sales, dtype: float64
```

Train Test Split

```
In [13]: from sklearn.model_selection import train_test_split
```

```
In [14]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size = 0.25, random_st
```

Model Training

```
In [15]: from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

```
In [16]: model = LinearRegression()
model.fit(x_train, y_train)
```

```
Out[16]:
```

LinearRegression ⓘ ?

Parameters

```
In [17]: print ("Intercept:", model.intercept_)
print ("coefficients:", model.coef_)
```

```
Intercept: 4.741842244124566
coefficients: [0.05461383 0.09975946 0.00430495]
```

Final Equation

$$y = 0.051 + 0.12 + 0.004 \cdot 3 + 4.71$$

In []: